

Forecasting nonstationary processes: what can be done?

Rebecca Killick (r.killick@lancs.ac.uk)
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- The challenge
- Tools required
- Forecasting
- Onward journey

What do I mean by Nonstationary Time Series?

- I mean **NOT** second-order stationary.
- So, unconditional variance **changes** with time.
- Autocovariance, spectrum, etc. **change** with time.

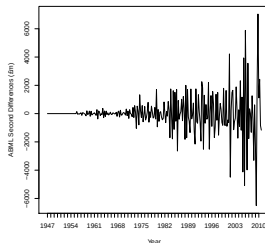
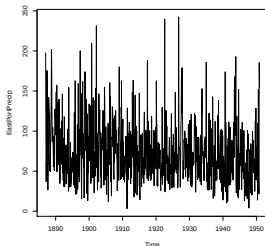
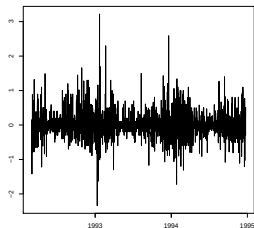
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More interested in Variability and CI



Time series diagnostics help to understand dynamics and inform modelling choices



Wave Heights, Precipitation, Economic

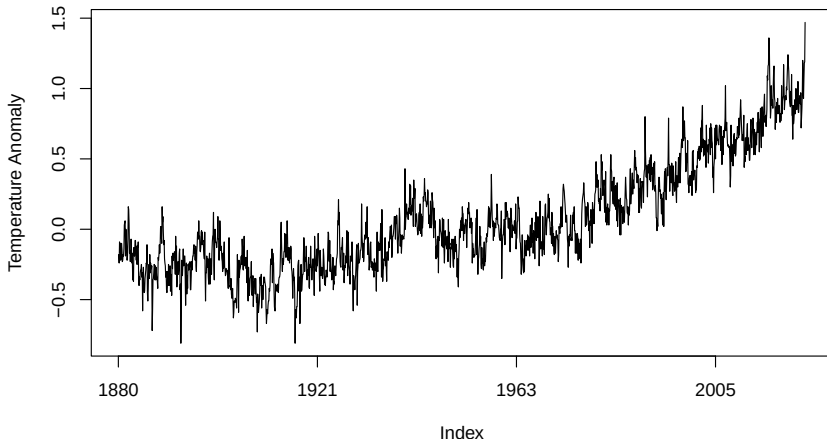


Stationary forecasting is hard to beat

Stationary forecasting is hard to beat

- Variance compensates for nonstationarity
- Can compensate with higher order models

GMST Example





- Non-stationary model
- Approach for forecasting the nonstationary structure
- Evaluation metric (based on variability NOT mean)

1. The model

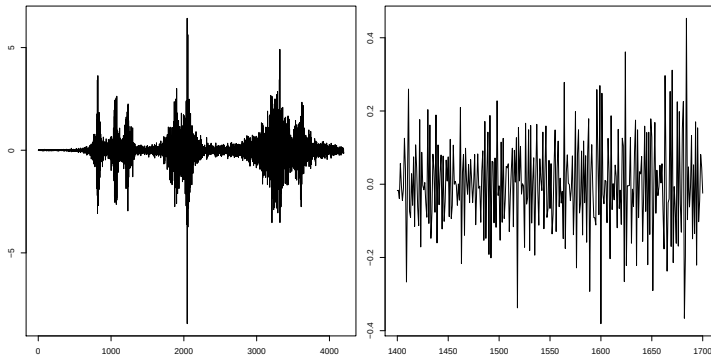
Modelling in the face of non-stationarity is no easy task!

- Various approaches have been explored, built on models that fit particular types of non-stationarity:
 - assume piecewise stationarity;
 - use parametric models with time-changing coefficients, \ e.g. Time Varying AR (tvAR).
- Processes with a slowly time-varying second order structure are known as **locally stationary** (LS).
 - Advanced LS models (ARCH) (Dahlhaus and Subba Rao, 2006).
 - Locally stationary fourier processes (Dahlhaus, 1997).
 - Locally stationary wavelet processes (Nason et al., 2000).

Locally stationary



If you take a small enough region, it will appear stationary as the structure varies slowly over time.



Application areas include;

- medicine, finance
- environmental processes, e.g. wind speeds.

A nonstationary time series can be modelled as a trend locally stationary wavelet (T-LSW) process $\{X_{t,T}\}$, $t = 0, \dots, T - 1$, as follows:

$$X_{t,T} = \mu(t/T) + \sum_{j=-\infty}^{-1} \sum_{k \in \mathbb{Z}} w_j(k/T) \psi_{j,k-t} \xi_{j,k},$$

where

- μ is a low order polynomial mean function,
- $w_j(k/T)$ are amplitudes, the smoothness of which controls the degree of nonstationarity,
- $\{\psi_{j,k-t}\}_{j,k}$ is a set of discrete non-decimated wavelets,
- $\{\xi_{j,k}\}$ is a sequence of orthonormal random variables.

McGonigle et al. (2022) JTSA, TLSW package on CRAN

The local autocovariance (LACV) is given by

$$c(z, \tau) = \sum_j S_j(z) \Psi_j(\tau),$$

- $\Psi_j(\tau)$ is the autocorrelation wavelet at scale j , lag τ
- $S_j(z)$ is the spectrum at rescaled time z

We estimate this using

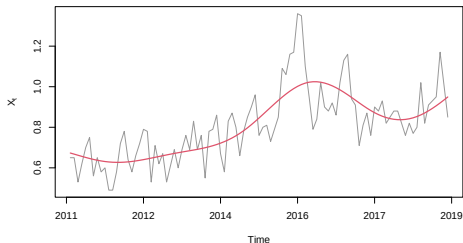
$$\hat{c}(z, \tau) = \sum_{j=-J_0}^{-1} \hat{S}_j(z) \Psi_j(\tau),$$

where $J_0 = \alpha \lfloor \log_2(T) \rfloor$ for $\alpha \in (0, 1)$ is the coarsest scale used.

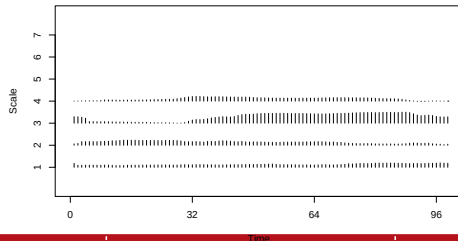
Example



Trend Estimate



Spectral Estimate



2. Forecasting approach

Assuming zero mean for a minute . . .

Suppose have data $x_1, \dots, x_T$ from **stationary** series.

Want to make forecast $\hat{x}_T(h)$ made at time T for horizon h .

Want forecast using linear combination of past:

$$\begin{aligned}\hat{x}_T(1) &= \sum_{i=0}^{T-1} \varphi_i x_{T-i} \\ &= \varphi_0 x_T + \varphi_1 x_{T-1} + \varphi_2 x_{T-2} + \dots\end{aligned}$$

Note: φ sequence DOES NOT depend on time (stationary x_t)

Theory can tell us optimal least-squares forecast (Box-Jenkins).

Stationary AR(p) model:

$$X_t = \sum_{i=1}^p \varphi_i X_{t-i} + \epsilon_t.$$

For forecasting we teach UG students to:

- Decide on the most appropriate model for the data;
 - use the pacf to aid choice of p ,
 - estimate the coefficients using MLE or Yule-Walker equations:

$$\begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_p \end{bmatrix} = \begin{bmatrix} \gamma_0 & \gamma_{-1} & \cdots \\ \gamma_1 & \gamma_0 & \cdots \\ \vdots & \vdots & \ddots \\ \gamma_{p-1} & \gamma_{p-2} & \cdots \end{bmatrix} \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \vdots \\ \varphi_p \end{bmatrix}$$

where $\gamma_{\cdot}$ is the autocovariance function for X_t .

- Forecast the data using the fitted coefficients and setting $\epsilon_t = 0$.

Given observations $x_0, \dots, x_{t-1}$, we:

- Predict $\hat{x}_t = \sum_{s=t-p}^{t-1} b_{t-1-i,T} X_s$, where
- p is the number of latest observations used for prediction and
- $\mathbf{b}$ is the solution to localized Yule-Walker equations.

$$\begin{bmatrix} c(t, 1) \\ c(t, 2) \\ \vdots \\ c(t, p) \end{bmatrix} = \begin{bmatrix} c(t-1, 0) & c(t-2, -1) & \cdots \\ c(t-1, 1) & c(t-2, 0) & \cdots \\ \vdots & \vdots & \ddots \\ c(t-1, p) & c(t-2, p-1) & \cdots \end{bmatrix} \begin{bmatrix} b_{t-1} \\ b_{t-2} \\ \vdots \\ b_{t-1-p} \end{bmatrix}$$

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These require knowledge of the covariance structure at time t which is the point we are trying to predict.

- The covariance at time t is similar to that at time $t-1$.
- We extrapolate at the smoothing step.

Thus we, mirror the stationary process

We need:

- To choose p appropriately
- To provide a data driven approach for practitioners

We define the lpacf at time t and lag τ as:

$$q(t, \tau) = \text{Corr}(X_t, X_{t-\tau} | \{X_{t-1}, \dots, X_{t-\tau+1}\})$$

$$q_T(t, \tau) = \begin{cases} \frac{c(\frac{t-1}{T}, 1)}{\sqrt{c(\frac{t}{T}, 0)c(\frac{t-1}{T}, 0)}}, & \text{for } \tau = 1 \\ \varphi_{t, \tau, \tau} \sqrt{\frac{\text{MSPE}(\hat{X}_t, X_t | X_{t-1}, \dots, X_{t-\tau+1})}{\text{MSPE}(\hat{X}_{t-\tau}, X_{t-\tau} | X_{t-1}, \dots, X_{t-\tau+1})}}, & \text{for } \tau \geq 2. \end{cases}$$

For stationary models the square root equals 1 and the $\varphi_{t, \tau, \tau}$ is the usual estimate of the pacf.

Theoretical treatment given in Killick et al. (2020) EJS, lpacf on CRAN

Sample convergence: As $T \rightarrow \infty$, uniformly in $\tau \in \mathbb{Z}$ and $z \in (0, 1)$,

$$|q_T(z, \tau) - q(z, \tau)| = \mathcal{O}(T^{-1}).$$

Estimate convergence: The lpacf estimate $\hat{q}(z, \tau)$ is consistent for the true lpacf, $q(z, \tau)$, i.e.

$$\hat{q}(z, \tau) \xrightarrow{P} q(z, \tau).$$

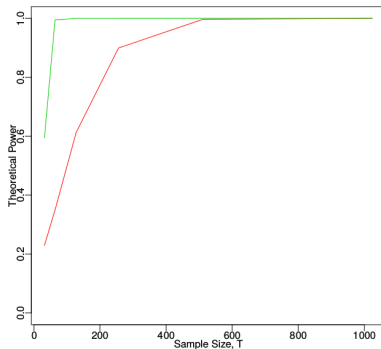
Window convergence: The windowed lpacf estimate $\hat{q}_W(z, \tau)$ is consistent for the true lpacf, $q(z, \tau)$, i.e.

$$\hat{q}_W(z, \tau) \xrightarrow{P} q(z, \tau).$$

Moreover, $\hat{q}_W(z, \tau)$ has a limiting Gaussian distribution.

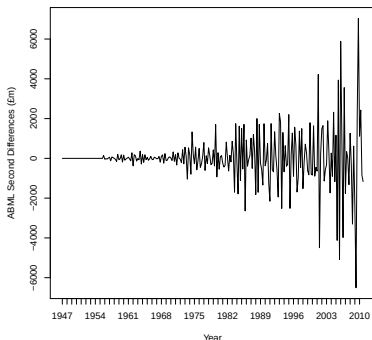
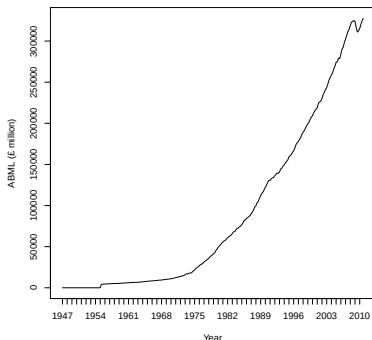


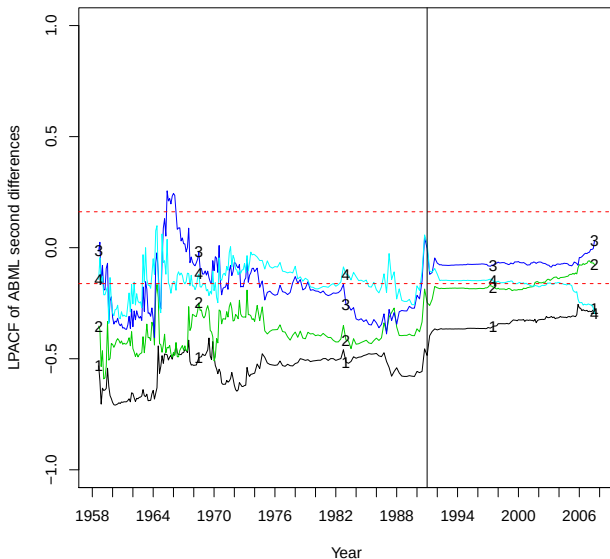
Interestingly we get a provably faster rate of convergence for the autocovariances than for the equivalent autocovariances from Locally Stationary Fourier processes.





- ABML consists of gross value added amounts
- Component in the estimate of GDP
- 223 observations from Q1 1955 to Q3 2010
- We use second differences to remove trend
- Tests of stationarity reject H_0 .





Thus we,

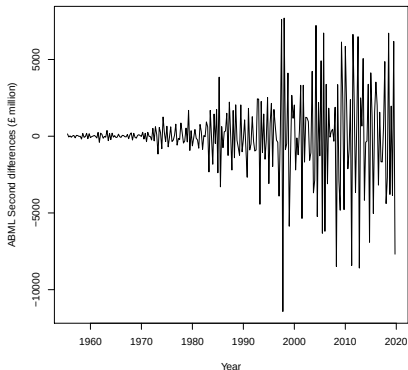
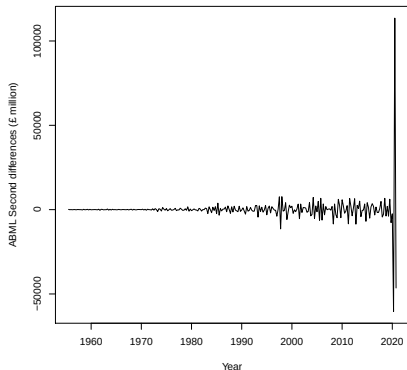
- mirror the stationary process
- build on the `ForecastLSW` approach
 - use the local partial autocorrelation function to choose p
- produce a data driven (with R package) approach for practitioners.

Range of models considered:

- TVAR(1)
- TVAR(2)
- TVAR(12)
- TVMA(1)
- TVMA(2)
- Uniformly modulated white noise
- LSW process

Summary

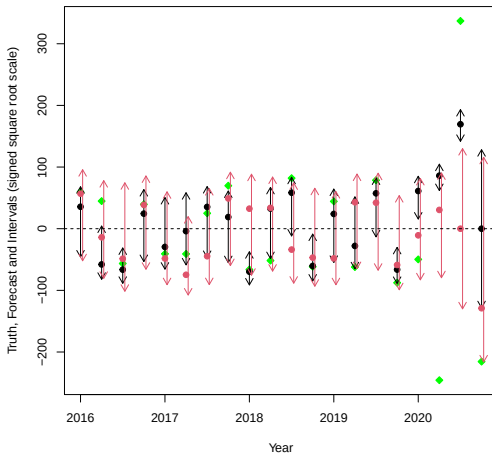
lpacf greatly improves on ARMA in 2/3 of cases
with comparable results (ratio ± 0.05) in remaining 1/3.





ForecastLSW package, Killick et al. JRSSC (2024+)

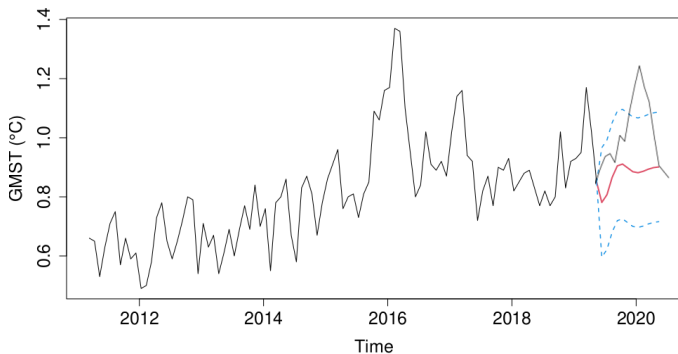
Black=B-J, Red=ForecastLSW





What about the mean structure?

Biggest challenge is boundary handling - Interval extension



- Non-stationary forecasting is hard
- ... but possible
- Recent advances in nonstationary methodology make forecasting possible
- Most methods are only doing point forecasts :-)
- Extensions to TLSW are showing good preliminary results
 - Now to theoretically back it up!

